Total No. of Questions :10]

SEAT No. :

P3744

[Total No. of Pages :3

[5561]-141
B.E. (Mechanical)
POWER PLANT ENGINEERING
(2012 Pattern) (Semester - II) (402047) (Theory)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3or Q.4, Q.5, or Q.6 Q.7or Q.8, Q.9 or Q.10.*
2) *Draw a neat diagram wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Use of calculator steam tables is allowed.*
5) *Assume suitable data, if necessary.*

Q1) a) What is meant by depreciation of power station? Explain the straight line method and sinking fund method of calculating depreciation. **[6]**

b) Explain the concept of Cascade Efficiency? **[4]**

OR

Q2) a) The maximum load on a thermal power plant of 60 MW capacity is 50 MW at an annual load factor of 60%. The coal consumption is 1 kg per unit of energy generated (kWh) and the cost of coal is Rs. 600 per ton of coal. Find: **[6]**

i) Annual revenue earned if the energy is sold at Rs. 1 per KW-hr.

ii) The capacity of the plant.

b) Define 'vacuum efficiency' applied to a condenser? State the effects of air leakage on the performance of a condenser? **[4]**

Q3) a) Explain with neat sketch pressurized Water Reactor (PWR). **[6]**

b) Explain with neat sketch types of spillways. **[4]**

OR

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Q4) a) Explain with neat sketch the working of modern steam power plant. **[6]**

b) Steam enters the condenser at 35.8°C with barometer reading of 760 mm of Hg. A vacuum of 700 mm was recorded in the condenser. Determine the vacuum efficiency. **[4]**

Q5) a) Enlist different applications of Diesel Power Plant. Explain Diesel Engine -Generating set (DG Set) size selection criteria for these applications. **[8]**

b) Derive the equation of thermal efficiency of Brayton cycle. **[8]**

OR

Q6) a) A open cycle gas turbine power plant operates on the Brayton cycle. The maximum pressure and temperature of the cycle are limited to 5 bar and 900 K. The pressure and temperature of the gas entering into the compressor are 1 bar and 27°C. Reheating is used at a pressure of 2.5 bar where the temperature of the gases is increased to its original turbine inlet temperature. The air flow per second through the plant is 10kg/sec. determine the thermal efficiency and plant capacity in MW. The exhaust pressure of the turbine is also 1 bar. Assume the compression and expansion are isentropic. Take γ =1.4 for air and gases. Cp=1KJ/kg-K for air and gases. And Calorific Value of the fuel is 40000KJ/kg. Neglect the pressure losses in the system. Do not neglect the fuel quantity. **[8]**

b) Why the supercharging is necessary in diesel plant? What are the methods used for supercharging the diesel engine? What are the advantages of supercharging? **[8]**

Q7) a) What are the Challenges in commercialization of Non-Conventional Power Plants? **[8]**

b) Write a short note on. **[8]**

i) Fuel Cell

ii) Bio-Gas Power plant

OR

Q8) a) Explain with neat sketch working of solar pond. **[8]**

b) What do you understand by MHD? Explain the working principle of MHD with neat sketches. **[8]**

Q9) a) Describe the methods used to control SO_2 and NO_x in steam power plant. **[9]**

b) Explain the layout of electrical equipment used in power plants. **[9]**

OR

Q10) a) What do you understand by thermal pollution? Explain the air and water pollution caused by thermal power plants. **[9]**

b) What is Thermal Discharge Index? Explain effects of Water & Noise pollution from power Plants & ways of reducing this pollution. **[9]**

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